

Rahul Anand Nayanegali

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SUMMARY

Senior Frontend Engineer with 6 years of experience building compliance-critical web applications using **React**, **TypeScript**, and **Micro Frontends (Module Federation)**. Specializes in schema-driven component architecture, micro-frontend migrations, and state management at production scale. Established TDD practices and AI-assisted **Playwright** pipelines that cut test authoring time 63% and raised E2E coverage 60%.

TECHNICAL SKILLS

Frontend: React 18, TypeScript, Next.js, Web Components, JavaScript, Core Web Vitals (LCP, INP, CLS), WCAG 2.2, State Management (Redux, Apollo Client)

Architecture: Micro Frontends (Module Federation), Monorepos (NX), Functional Programming, TDD, Clean Architecture, SOLID

AI-Augmented Engineering: Claude Code, Kiro, MCP, OpenCode CLI; prompt engineering (CoT, Few-Shot) for accelerated delivery

Testing: React Testing Library, Jest, Playwright, Cypress

Backend: Python, Flask, Node.js, REST APIs, GraphQL, WebSockets, SSE (Real-Time)

Cloud & Tools: AWS (ECS, EC2, S3, Lambda, CloudFront), Docker, Chrome Extensions, Vite/Webpack, NX, Git, CI/CD

Certifications: AWS Certified Cloud Practitioner; Master JavaScript – LinkedIn Learning

EXPERIENCE

Senior Frontend Engineer

Jun 2025 – Dec 2025

Amazon (Contract via Lares IT Solutions Inc)

Seattle, WA

- Migrated a business-critical **React/TypeScript** frontend from legacy architecture to **micro-frontends (Module Federation)** using the **Strangler Fig** pattern; maintained zero production downtime across the incremental cutover.
- Built and evolved schema-driven UI modules in **React** as a primary contributor to **ActionWidget**, a micro-frontend engine powering dynamic compliance workflows for Amazon sellers globally.
- Implemented **TDD** and trunk-based development practices; stabilized production releases and reduced deployment cycle time across the frontend team.
- Designed AI-assisted **Playwright** test authoring pipeline using BDD specifications as LLM context, cutting test authoring time 63%, raising E2E coverage 60%, and accelerating delivery velocity 30%.
- Raised code review quality by enforcing single-purpose diffs (structural refactors separated from behavioral changes), reducing cognitive load and integration risk during active Module Federation migration.

Senior Software Engineer

Aug 2024 – Jun 2025

Shoptaki

New York, NY

- As the sole frontend engineer during stealth phase, owned end-to-end delivery of three core product surfaces: plain-English-to-SQL query interface, developer dashboard, and user authentication flows.
- Integrated an LLM with live database schema as context to power a natural-language SQL interface; reduced query turnaround from hours to seconds for non-technical team members.
- Engineered developer dashboard from scratch serving the core engineering team across logs, metrics, and user management; built Firebase authentication API handling password change and account recovery end-to-end.

Engineer, Web Development

Feb 2021 – Jan 2023

CodeCraft Technologies

Bangalore, India

- Co-owned **React/TypeScript** component development for JioCinema on **Next.js 13** (Pages Router) with **Redux**; implemented **SSR** and code splitting via dynamic imports on streaming pages serving IPL and FIFA World Cup 2022 traffic.
- Modernized build pipeline for a high-traffic streaming platform (Webpack/Babel upgrade, asset optimization); cut build times 50% and unblocked ES6+ feature adoption.
- Migrated application state from **Redux** to **Apollo Client (GraphQL)**, implementing a unified GraphQL infrastructure layer; eliminated boilerplate and reduced bundle size 20%.
- Built and maintained backend services in **Node.js** and Python on **AWS** (ECS, EC2, Lambda, S3); engineered robust data processing pipelines.

Assistant Systems Engineer

Nov 2018 – Apr 2020

Tata Consultancy Services

Bangalore, India

- Built real-time IoT monitoring dashboard using **Angular**, Flask, and **Server-Sent Events (SSE)**, handling continuous sensor data streams at scale; designed RESTful API endpoints for auth, data retrieval, and live updates.
- Optimized frontend performance via lazy loading, image compression, and JS minification; delivered measurable load time improvements on production dashboard.

EDUCATION

Pace University

New York, NY

Master of Science in Computer Science

Jan 2023 – May 2024

Sri Siddhartha Institute of Technology

Tumkur, India

Bachelor of Engineering in Electronics and Communication Engineering

September 2018